

PHYS 230 - Electricity and Magnetism

Lecture 11

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

3 Electric Potential Energy and Potential

- Electric Potential Energy (Continued)

All figures in these slides are from one of the following sources, unless cited otherwise:

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Physics for Scientists and Engineers, Hawkes, et al.

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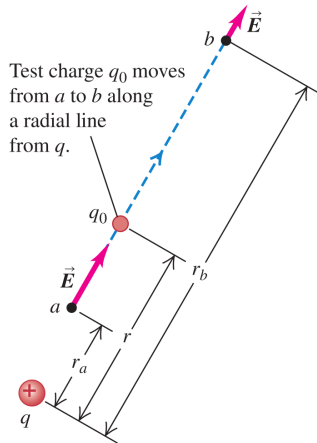
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Example: Electric Potential Energy of Two Point Charges

- Take charge q and a test charge q_0

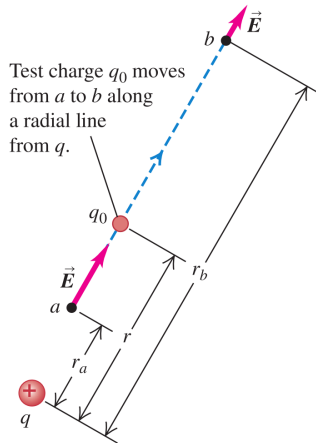
Figure 23.5 Test charge q_0 moves along a straight line extending radially from charge q . As it moves from a to b , the distance varies from r_a to r_b .



Example: Electric Potential Energy of Two Point Charges

- Take charge q and a test charge q_0
- Move q_0 from point $\mathbf{r}_a$ to point $\mathbf{r}_b$ in the electric field of q , on a radial path connecting q and q_0

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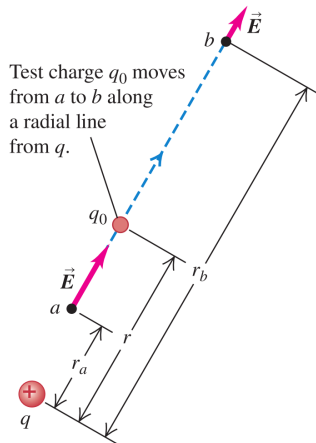
- Take charge q and a test charge q_0
- Move q_0 from point $\mathbf{r}_a$ to point $\mathbf{r}_b$ in the electric field of q , on a radial path connecting q and q_0
- The change in potential energy of the system

$$\begin{aligned}U_b - U_a &= -q_0 \int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{E} \cdot d\mathbf{r} = -q_0 \int_{\mathbf{r}_a}^{\mathbf{r}_b} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \right) \cdot (dr \hat{\mathbf{r}}) \\&= -q_0 \int_{\mathbf{r}_a}^{\mathbf{r}_b} \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr \underbrace{\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}}_1 \\&= -\frac{q_0 q}{4\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r^2} = -\frac{q_0 q}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_{r_a}^{r_b}\end{aligned}$$

or

$$U_b - U_a = \frac{q_0 q}{4\pi\epsilon_0} \left(\frac{1}{r_b} - \frac{1}{r_a} \right) = \underbrace{\frac{1}{4\pi\epsilon_0} \frac{q_0 q}{r_b}}_{U_b} - \underbrace{\frac{1}{4\pi\epsilon_0} \frac{q_0 q}{r_a}}_{U_a}$$

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Electric Potential Energy

To find the potential energy, not the potential energy difference, of a configuration

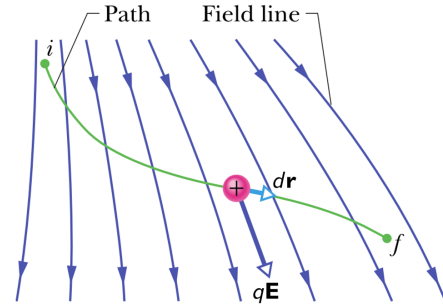


Fig. 24-4 A test charge q moves from point i to point f along the path shown in a nonuniform electric field. During a displacement $d\mathbf{r}$, an electrostatic force $q\mathbf{E}$ acts on the test charge. This force points in the direction of the field line at the location of the test charge.

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1. Choose the initial (reference) point i of all the involved charges somewhere where the potential energy between them is zero: this is usually arbitrary and is taken to be at infinity

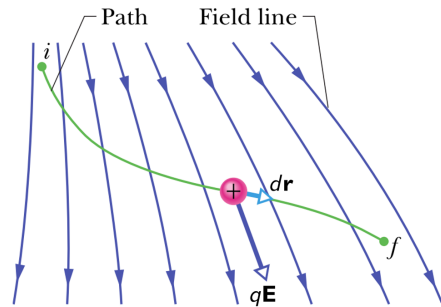


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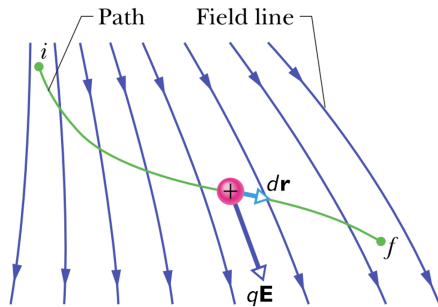


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2. Choose the final points of all charges as their position in the configuration for which you want to find the potential energy
3. Then we take the $U_f = U$ as the potential energy of the configuration

$$\underbrace{U_f}_U - \underbrace{U_i}_{U_{\text{ref}}=0} = -q \int_{\mathbf{r}_i=\mathbf{r}_{\text{ref}}}^{\mathbf{r}_f} \mathbf{E} \cdot d\mathbf{r}$$

or

$$U = -q \int_{\mathbf{r}_i=\mathbf{r}_{\text{ref}}}^{\mathbf{r}_f} \mathbf{E} \cdot d\mathbf{r}$$

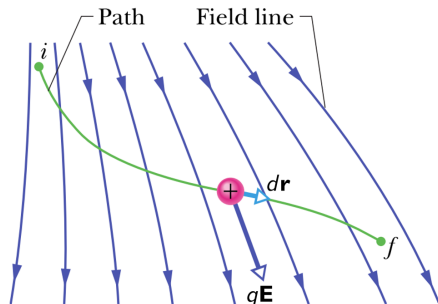


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- Potential energy is always defined relative to some reference point where $U = 0$

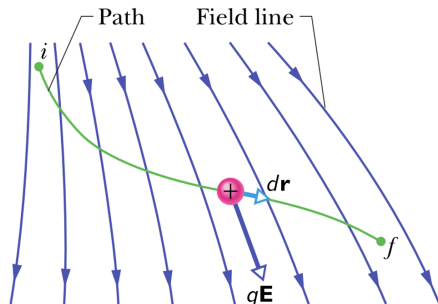


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Important Point!

- In any computation which involves an integral over a curve with the infinitesimal displacement $d\mathbf{r}$, regardless of the direction of the curve, DO NOT CHANGE THE DIRECTION (SIGN) OF $d\mathbf{r}$.
- **Change of integral limits** due to change of direction of movement on the curve **will take care of the sign!**

Example: Electric Potential Energy of Two Point Charges

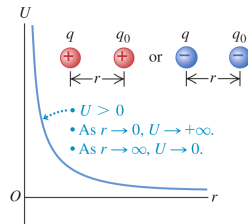
What is the potential energy of a configuration where two point charges q , q_0 are placed at a distance r from each other?

We take r_a at infinity and bring q_0 from infinity to $r_b = r$, then the potential energy of the configuration where q and q_0 have a distance of r from each other is

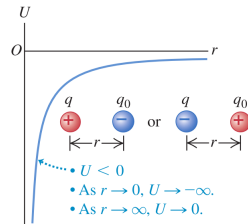
$$\underbrace{U_b}_{U} - \underbrace{U_a}_0 = \frac{q_0 q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{\infty} \right) = \frac{1}{4\pi\epsilon_0} \frac{q_0 q}{r}$$

Figure 23.7 Graphs of the potential energy U of two point charges q and q_0 versus their separation r .

(a) q and q_0 have the same sign.



(b) q and q_0 have opposite signs.



Exercise I

<https://epoll.srv.ualberta.ca> – Code: MBD

A positron (electron's antiparticle) has mass 9.11×10^{-31} kg and charge $q_p = +e = +1.60 \times 10^{-19}$ C. Suppose a positron moves in the vicinity of an α particle, which has charge $q_\alpha = +2e = 3.20 \times 10^{-19}$ C and mass 6.64×10^{-27} kg. The α particle's mass is more than 7000 times that of the positron, so we assume that the α particle remains at rest. When the positron is 1.00×10^{-10} m from the α particle, it is moving directly away from the α particle at 3.00×10^6 m/s.

- (a) What is the positron's speed when the particles are 2.00×10^{-10} m apart?
- (b) What is the positron's speed when it is very far from the α particle?

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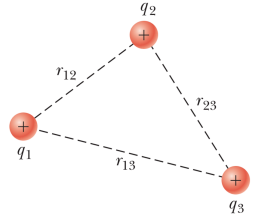
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Electric Potential Energy of A System of Point Charges

What is the potential energy of a configuration where several point charges $q_1, q_2, \dots, q_n$ are at $\mathbf{r}_1, \mathbf{r}_2, \dots$, etc.?



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- Remove all charges from their positions and keep them infinitely apart.

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- At this point, $U_{\text{total}} = 0$, because there are no charges around.

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- Bring in q_1 from $r = \infty$ to $\mathbf{r}_1$.



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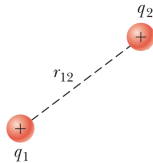
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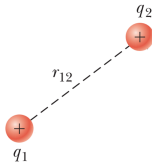
- Bring in q_2 from $r = \infty$ to $\mathbf{r}_2$.



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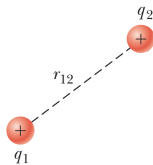
- Bring in q_2 from $r = \infty$ to $\mathbf{r}_2$.
- Charge q_2 will feel electric force $\mathbf{F}_{12} = q_2 \mathbf{E}_1$ from q_1



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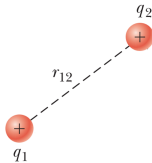
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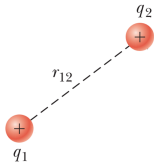
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- System will have a potential energies U_{12}
- Total potential energy of the system is now

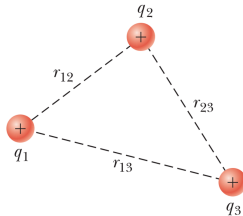


$$U_{\text{total}} = U_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

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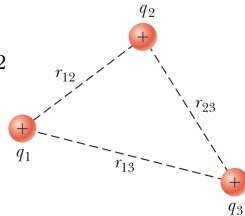
- Bring in q_3 from $r = \infty$ to $\mathbf{r}_3$.



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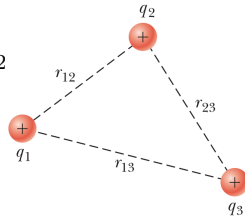
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- It will feel electric forces $\mathbf{F}_{13} = q_3 \mathbf{E}_1$ and $\mathbf{F}_{23} = q_3 \mathbf{E}_2$ from both q_1 and q_2



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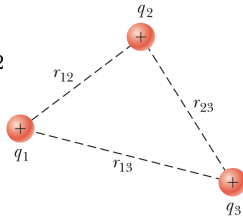
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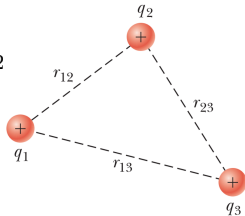
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- System will have two additional potential energies $U_{13} + U_{23}$
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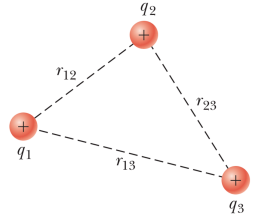


$$U_{\text{total}} = U_{12} + U_{13} + U_{23} = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$$

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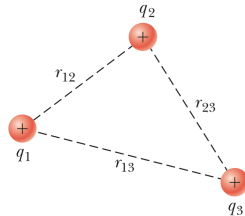


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- The U_{total} **is the sum of U of pairs**, i.e.,

$$U_{\text{total}} = \frac{1}{2} \sum_{\substack{i,j=1 \\ i \neq j}}^n U_{ij} = \frac{1}{2} \left[\frac{1}{4\pi\epsilon_0} \sum_{\substack{i,j=1 \\ i \neq j}}^n \frac{q_i q_j}{r_{ij}} \right]$$



Exercise 2

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Two point charges are at fixed positions on the x -axis, $q_1 = -e$ at $x = 0$ and $q_2 = +e$ at $x = a$.

(a) Find the work that must be done by the field of q_1 and q_2 to bring a third point charge $q_3 = +e$ from infinity to $x = 2a$.

(b) Find the total potential energy of the system of three charges.

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